

DATA ANALYTICS, DASHBOARD & REPORTING SERVICE

Professional Service Delivery Guideline

From raw data to clear decisions

Prepared by: Chala Tesfaye

MEAL Manager

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Table of Contents

1. Purpose	3
2. Service Objectives	3
Professional Delivery Principles	3
3. Service Delivery Process	4
4. Phase 1 — Requirements Assessment	4
5. Phase 2 — Data Collection & Data Inventory	5
6. Phase 3 — Data Quality Assessment	5
7. Phase 4 — Data Cleaning	7
8. Phase 5 — Data Transformation	7
9. Phase 6 — KPI & Indicator Development	8
10. Phase 7 — Data Analysis	9
11. Phase 8 — Dashboard Development	9
12. Phase 9 — Reporting Automation	10
13. Phase 10 — Data Validation & Quality Assurance	11
14. Phase 11 — Management Insights	12
15. Phase 12 — Report Development	13
16. Phase 13 — Documentation	14
17. Phase 14 — Client Review	14
18. Phase 15 — Final Handover	15
19. Post-Delivery Support	16
20. Client Responsibilities	16
21. Data Confidentiality	17
22. Standard Deliverables	17
23. Technology	18
24. Success Criteria	18
25. Final Service Output	19

1. Purpose

This service helps organizations turn raw data into clear, reliable, and decision-ready information. It brings together careful data preparation, practical analysis, and accessible reporting so that teams can understand performance and act with confidence.

The work covers the full journey: understanding the client's needs, assessing and preparing the data, developing meaningful indicators, building dashboards and reports, checking quality, and handing over a solution that people can use and maintain.

2. Service Objectives

The service aims to:

- ✚ Improve data quality and consistency.
- ✚ Establish reliable and standardized datasets.
- ✚ Reduce manual data-processing and reporting work.
- ✚ Develop clear and relevant performance indicators.
- ✚ Produce interactive dashboards and reports.
- ✚ Establish repeatable reporting processes.
- ✚ Improve management access to timely information.
- ✚ Identify trends, gaps, variations, and performance issues.
- ✚ Support evidence-based decision-making.
- ✚ Provide documented and maintainable reporting solutions.

Professional Delivery Principles

Every assignment will be guided by a few practical principles that keep the work useful, responsible, and focused on the people who will use it.

- Clarity first: present findings in language that decision-makers and operational teams can understand.
- Traceability: keep a clear link between source data, transformations, calculations, and final results.
- Proportionality: use methods and tools that match the client's needs, capacity, budget, and risk.

- User involvement: validate requirements and outputs with the people who will use the solution.
- Responsible data use: protect confidentiality, minimize unnecessary data, and apply agreed access controls.
- Sustainability: design processes that the client can refresh, explain, and maintain after handover.
- Inclusive access: design outputs for different levels of technical confidence and, where practical, consider accessibility and low-bandwidth use.
- Communication and adoption: explain changes early, record key decisions, and provide guidance that helps users adopt the new reporting process.

3. Service Delivery Process

The service follows a clear, collaborative workflow. Each stage builds on the previous one, with client feedback and quality checks included throughout rather than left until the end.

Client Requirements → Data Assessment → Data Preparation → Data Quality Assurance → Data Transformation → Data Modeling → KPI Development → Data Analysis → Dashboard Development → Reporting Automation → Validation → Insights → Documentation → Handover → Support

4. Phase 1: Requirements Assessment

Before any data work begins, the consultant and client will agree on the decisions the analysis should support, the people who will use the outputs, and the practical requirements for reporting.

Activities

- ✚ Identify the purpose of the analysis.
- ✚ Identify the intended users of the output.
- ✚ Identify key business/program questions.
- ✚ Identify reporting frequency.
- ✚ Identify required indicators and KPIs.
- ✚ Identify required reporting periods.
- ✚ Identify required breakdowns and dimensions.

- ✚ Review existing reporting processes.
- ✚ Review existing dashboards and reports where applicable.
- ✚ Identify required outputs and delivery formats.
- ✚ Establish data-security and confidentiality requirements.

Output

Requirements and Reporting Specification

This document defines what information will be analyzed, how it will be presented, and what will be delivered.

5. Phase 2: Data Collection & Data Inventory

The next step is to build a shared picture of the available data. Relevant sources will be gathered, documented, and checked for ownership, coverage, format, frequency, and relationships.

Activities

- ✚ Collect available datasets.
- ✚ Identify data sources.
- ✚ Identify data owners.
- ✚ Identify reporting periods.
- ✚ Identify data formats.
- ✚ Identify data frequency.
- ✚ Identify relationships between datasets.
- ✚ Identify missing datasets or reporting periods.
- ✚ Establish the source-of-truth dataset where applicable.

Output

Data Inventory and Source Register

6. Phase 3: Data Quality Assessment

Before analysis starts, the supplied data will be reviewed systematically to understand what is reliable, what needs attention, and what limitations should be communicated.

Quality dimensions

- Completeness
- Accuracy
- Consistency
- Validity
- Uniqueness
- Timeliness
- Integrity

Assessment activities

- ✚ Identify missing values.
- ✚ Identify duplicate records.
- ✚ Identify inconsistent values.
- ✚ Identify invalid values.
- ✚ Check data types.
- ✚ Check date fields.
- ✚ Check categorical variables.
- ✚ Check numerical values.
- ✚ Check identifiers.
- ✚ Check reporting periods.
- ✚ Identify outliers requiring investigation.
- ✚ Identify structural problems.

Output

Data Quality Assessment Report

The report will document identified issues and required corrective actions.

7. Phase 4: Data Cleaning

Based on the agreed rules, the data will be cleaned and standardized in a controlled way. Changes will be documented so that the process remains transparent and repeatable.

Activities

- Remove or resolve duplicate records.
- Standardize field names.
- Standardized categories.
- Correct formatting inconsistencies.
- Standardized dates.
- Standardize numerical fields.
- Address missing values according to agreed rules.
- Remove irrelevant fields where appropriate.
- Correct identifiable data-entry inconsistencies.
- Maintain data integrity and traceability.

Output

Cleaned and Standardized Dataset

The original source data will remain preserved separately from the cleaned dataset.

8. Phase 5: Data Transformation

Once cleaned, the data will be reshaped into a practical structure for analysis and reporting, with clear fields, consistent dimensions, and appropriate relationships between datasets.

Activities

- Combine datasets where required.
- Append datasets where required.

- Merge related datasets.
- Create required fields.
- Create calculated fields.
- Structure dates and reporting periods.
- Standardize dimensions.
- Prepare analytical tables.
- Establish appropriate relationships between datasets.

Output

Analysis-Ready Dataset / Data Model

9. Phase 6: KPI & Indicator Development

Together with the client, the required KPIs and indicators will be defined in plain language and documented well enough for different users to calculate and interpret them consistently.

For each KPI/indicator, the following will be established where applicable:

- Indicator name
- Definition
- Purpose
- Data source
- Calculation method
- Numerator
- Denominator
- Unit of measurement
- Target
- Actual value
- Reporting frequency
- Reporting period

- Responsible data source/owner
- Disaggregation requirements

Output

KPI / Indicator Dictionary

10. Phase 7: Data Analysis

The prepared dataset will be analyzed against the agreed questions and indicators. The focus will be on explaining patterns, differences, progress, and exceptions: not simply producing numbers.

Analysis areas

- ✚ Descriptive analysis
- ✚ Trend analysis
- ✚ Comparative analysis
- ✚ Performance analysis
- ✚ Variance analysis
- ✚ Target versus actual analysis
- ✚ Period-to-period analysis
- ✚ Category analysis
- ✚ Geographic analysis
- ✚ Departmental analysis
- ✚ Outlier analysis
- ✚ Data-quality analysis

Output

Data Analysis Results

11. Phase 8: Dashboard Development

An interactive dashboard will be developed around the approved questions and KPI framework. Its design will prioritize clear navigation, readable visuals, and the information users need most often.

Dashboard components may include

- ✚ KPI cards
- ✚ Performance indicators
- ✚ Trend charts
- ✚ Comparison charts
- ✚ Tables
- ✚ Summary views
- ✚ Detailed views
- ✚ Filters
- ✚ Slicers
- ✚ Date controls
- ✚ Drill-down functionality
- ✚ Conditional formatting
- ✚ Performance status indicators

The dashboard will be designed for usability, clarity, consistency, and management decision-making.

Output

Interactive Dashboard

12. Phase 9: Reporting Automation

Where the data environment allows, recurring reporting steps will be simplified or automated. The aim is to reduce avoidable manual effort while keeping the process easy to review and control.

Activities

- Establish standardized data-input structures.
- Establish repeatable data-transformation processes.
- Automate recurring calculations.
- Automate dashboard updates where technically feasible.
- Establish refresh procedures.
- Standardize recurring reporting outputs.
- Reduce manual copying and recalculation.

Target workflow

New Data → Data Refresh → Updated Analysis → Updated Dashboard → Updated Report

Output

Repeatable / Automated Reporting System

13. Phase 10: Data Validation & Quality Assurance

Before delivery, the data, calculations, dashboards, and refresh process will be tested against agreed checks. Any exceptions will be resolved or clearly documented.

Validation areas

Data validation

- Source data reconciliation
- Record counts
- Totals
- Missing values
- Duplicate records

- Data consistency

Calculation validation

- KPI formulas
- Aggregations
- Percentages
- Ratios
- Growth calculations
- Target calculations

Dashboard validation

- Filters
- Slicers
- Dates
- Charts
- KPI values
- Tables
- Navigation
- Drill-down functionality

Refresh validation

- Data refresh
- Transformation steps
- Calculations
- Dashboard updates

Output

Validated Reporting Solution

14. Phase 11: Management Insights

Validated results will be translated into a concise management narrative that explains what has changed, why it matters, and where attention may be needed.

The analysis may identify:

- Significant trends
- Performance changes
- Variations
- Gaps
- Exceptions
- Underperforming areas
- High-performing areas
- Data-quality concerns
- Areas requiring further investigation

The purpose of this stage is to make the analytical results understandable and useful for decision-making.

Output

Management Insights Summary

15. Phase 12: Report Development

When a formal report is required, it will bring the validated evidence, interpretation, and supporting visuals into one coherent document suited to the intended audience.

The report may include:

1. Executive summary
2. Methodology
3. Data overview
4. KPI results
5. Performance analysis

6. Trend analysis
7. Key findings
8. Data-quality findings
9. Management considerations
10. Supporting tables/charts

Output

Final Analytical / Performance Report

16. Phase 13: Documentation

The solution will be documented so that future users can understand how the data flows, how indicators are calculated, and how reports can be refreshed and maintained.

Documentation may include:

- Data dictionary
- KPI dictionary
- Data-cleaning rules
- Transformation procedures
- Calculation methodology
- Dashboard guide
- Data-refresh procedure
- Reporting procedure
- Technical documentation

Output

Reporting System Documentation

17. Phase 14: Client Review

The completed solution will be shared with the client for a practical review. Feedback will be checked against the agreed scope, documented, and addressed before approval.

The client will review:

- Data accuracy
- KPI definitions
- Dashboard functionality
- Report content
- Required modifications
- Usability
- Reporting requirements

Agreed revisions will be incorporated before final handover.

Output

Client-Approved Reporting Solution

18. Phase 15: Final Handover

After approval, the final solution will be handed over in the agreed format, together with the files and guidance needed for confident day-to-day use.

Depending on the project, the final package may include:

- Clean dataset
- Data model
- Excel dashboard
- Power BI dashboard
- KPI/indicator dictionary
- Data-quality report
- Analytical report
- Management insights

- User guide
- Refresh instructions
- Technical documentation

19. Post-Delivery Support

After handover, limited support may be provided to help the client resolve early issues, apply agreed minor updates, and build confidence in maintaining the solution.

- Dashboard troubleshooting
- Data-refresh support
- Minor corrections
- Reporting updates
- KPI modifications
- Data-source changes
- User guidance
- Technical support

The duration and scope of support will be agreed with the client before project commencement.

20. Client Responsibilities

Successful delivery depends on timely collaboration. The client will be responsible for:

- Providing relevant data.
- Providing accurate source information.
- Identifying authorized data sources.
- Confirming KPI definitions.
- Providing required reporting requirements.
- Reviewing outputs within the agreed timeframe.
- Confirming corrections or changes.

- Providing appropriate access where required.
- Ensuring that the data supplied can legally be processed.

21. Data Confidentiality

Client data will be handled with care, treated as confidential, and used only for the agreed analytical and reporting purpose.

Access will be limited to the information and people genuinely required to deliver the service.

Before work begins, the client and service provider will agree how data will be stored, transferred, retained, and securely disposed of, taking account of its sensitivity and applicable requirements.

22. Standard Deliverables

The final deliverables will be determined according to the project scope.

The standard service can include:

Deliverable	Description
Data Assessment	Review of data and reporting requirements
Data Quality Report	Identified data-quality issues
Clean Dataset	Standardized analysis-ready data
Data Model	Structured analytical data
KPI Dictionary	Definitions and calculation methodology
Dashboard	Interactive Excel/Power BI dashboard
Analysis	Statistical and performance analysis
Automated Reporting	Repeatable reporting workflow
Management Insights	Key findings from the data
Analytical Report	Formal reporting output

Deliverable	Description
User Guide	Instructions for using the solution
Refresh Guide	Instructions for updating the data
Technical Documentation	Documentation of the reporting system

23. Technology

The service may utilize:

- Microsoft Excel
- Power Query
- Power Pivot
- Power BI
- DAX
- SQL
- Statistical analysis tools
- Data visualization tools
- Appropriate automation and AI technologies where beneficial

Technology choices will be made with the client's data environment, user capacity, budget, security needs, and reporting objectives in mind. The simplest dependable option will be preferred over unnecessary complexity.

24. Success Criteria

The solution will be considered successful when users can trust the results, understand how they were produced, and maintain the reporting process with reasonable effort. In practical terms, it should provide:

- Reliable and structured data
- Clearly defined KPIs

- Accurate calculations
- Interactive reporting
- Reduced repetitive manual work
- Repeatable reporting processes
- Clear management information
- Documented methodology
- Maintainable reporting processes
- Appropriate data-quality controls

25. Final Service Output

The overall service transforms:

RAW DATA

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DATA QUALITY ASSESSMENT

↓

CLEAN DATA

↓

DATA TRANSFORMATION

↓

DATA MODEL

↓

KPI FRAMEWORK

↓

DATA ANALYSIS

↓

DASHBOARD

↓

REPORTING AUTOMATION

↓

QUALITY ASSURANCE

↓

MANAGEMENT INSIGHTS

↓

FINAL REPORT

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DOCUMENTATION & HANDOVER

The goal is not simply to deliver a spreadsheet or visualization. It is to leave the client with a **complete, accurate, practical, repeatable, and maintainable data analytics and reporting solution**: one that people trust, understand, and can use to make better decisions.